

Title page

Title A liver cancer lineage axis and LAT1 vulnerability reveal non-identifiability in single-cohort proteogenomic merges

Running title Liver cancer lineage axis and LAT1 vulnerability

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Dear Editor-in-Chief,

Please consider the manuscript, “A liver cancer lineage axis and LAT1 vulnerability reveal non-identifiability in single-cohort proteogenomic merges,” for publication as an Original Research Article in Translational Oncology.

The study fits the journal’s translational oncology scope because it connects cancer lineage classification, target nomination, and clinical anchoring in liver cancer. Using only open data, it builds

a cross-validated hepatocellular-carcinoma (HCC) to intrahepatic-cholangiocarcinoma (iCCA) lineage axis (AUC 0.966) and places combined hepatocellular-cholangiocarcinoma (cHCC-CCA) in the expected intermediate direction, with independent replication relative to HCC.

The paper also addresses a practical target-nomination risk. An apparent iCCA-selective EGFR/kinase program disappears when DepMap dependency is tested against HCC rather than all other lineages. By contrast, LAT1 amino-acid transport remains supported by tumor expression and by the positive randomized phase 2 nanvuranlat study in biliary tract cancer.

The methodological contribution is a reproducible benchmark showing that single-cohort-per-tumor-type proteogenomic merges are non-identifiable for cross-tumor contrasts. A standard harmonization produced thousands of liver and kidney protein hits that failed batch-free RNA validation, and ComBat could not resolve the aliased design. The manuscript therefore separates what open liver cancer data can support from what they cannot.

All data are public, and the complete pipeline, tables, and figures are released on GitHub (<https://github.com/Goden26/single-cohort-proteogenomic-nonidentifiability>) and archived at Zenodo (DOI: 10.5281/zenodo.20888892).

Thank you for considering this submission.

Sincerely,

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Highlights

- A cross-validated HCC-iCCA lineage axis places cHCC-CCA as intermediate.
- Single-cohort proteogenomic merges fail under perfect cohort-tumor aliasing.
- The correct DepMap comparator removes an apparent EGFR dependency signal.
- LAT1 amino-acid transport remains a clinically anchored iCCA vulnerability.

A liver cancer lineage axis and LAT1 vulnerability reveal non-identifiability in single-cohort proteogenomic merges

Abstract

Background. Hepatocellular carcinoma (HCC) and intrahepatic cholangiocarcinoma (iCCA) form a lineage continuum, with combined hepatocellular-cholangiocarcinoma (cHCC-CCA) intermediate. Using only open data, we sought liver cancer resources for lineage classification and target nomination, and tested whether single-cohort proteogenomic merges support cross-tumor conclusions.

Methods. From TCGA-LIHC (n=371) and TCGA-CHOL (n=35) RNA-seq we built an HCC-to-iCCA lineage axis (nested cross-validation, permutation, non-marker hold-out), positioned cHCC-CCA, and replicated the direction. We tested Gao 2019 HCC and Dong 2022 iCCA

proteome/phosphoproteome against a batch-free RNA anchor, ComBat, ground-truth titration, a kidney pair, and DepMap dependency using the direct iCCA-vs-HCC comparator.

Results. The axis was robust (cross-validated AUC 0.966); cHCC-CCA shifted toward iCCA in two cohorts. An apparent iCCA-selective kinase program was a comparator artifact: EGFR was neither RNA-concordant (log2FC 0.38) nor differentially essential (FDR 0.63), while FGFR2/IDH1 remain the validated programs. LAT1 amino-acid transport survived, was over-expressed in iCCA tumors (log2FC +0.79; $P = 1.2e-10$), and aligned with phase-2 nanvuranlat data in biliary tract cancer (HR 0.56, $P = 0.02$). The single-cohort proteome did not recover lineage biology ($\rho = -0.15$); naive harmonization flagged 63% of liver and 76% of kidney proteins, none validating.

Conclusions. Open transcriptomes provide a reusable liver cancer lineage axis and a clinically anchored LAT1 signal. Condition-selective dependencies need the correct comparator because single-cohort-per-type proteogenomic merges are non-identifiable for cross-tumor contrasts.

Keywords: Hepatocellular carcinoma; Intrahepatic cholangiocarcinoma; Combined HCC-CCA; LAT1; Cancer dependency; Proteogenomics; Batch confounding

Introduction

Hepatocellular carcinoma (HCC) and intrahepatic cholangiocarcinoma (iCCA) are the two dominant primary liver cancers, and combined hepatocellular-cholangiocarcinoma (cHCC-CCA) is a rare biphenotypic entity whose taxonomy was standardized only recently [14]. Single-cell and spatial work has reframed the HCC to iCCA distinction as a continuum of lineage states. That creates two practical needs: a robust way to place tumors on the continuum and a reliable way to nominate lineage-relevant targets. Using only open data, we provide a cross-validated HCC-to-iCCA transcriptomic axis, show how an apparent EGFR/iCCA dependency dissolves under the correct comparator, and recover LAT1 amino-acid transport as a modest but clinically anchored vulnerability.

A tempting shortcut is to harmonize the two largest open liver proteogenomic cohorts into one protein- and phosphosite-level manifold and read off lineage-changing activities and dependencies. Gao and colleagues characterized 159 HBV-related HCCs with paired proteome and phosphoproteome [1], and Dong and colleagues an iCCA cohort with the same layers [2]. The obstacle is structural: each tumor type is one cohort, processed in one laboratory with one mass-spectrometry reference, so tumor type is perfectly aliased with batch, and for HCC also with HBV etiology. Under perfect aliasing no procedure can separate the biological contrast from the technical one. The merge is non-identifiable for the cross-tumor contrast; the established safeguard is to validate any cross-cohort contrast against an independent batch-free reference and distrust low concordance [21,22,23,24].

We tested the merge rather than assuming it could work. We established the lineage axis from a batch-controlled layer (TCGA-LIHC and TCGA-CHOL RNA-seq on the identical GDC pipeline), validated it out of sample, positioned cHCC-CCA and replicated the direction, then asked whether the single-cohort proteome and phosphoproteome recover the same biology. Because they do not, we used DepMap CRISPR dependency, a single-platform validation layer, to test lineage-selective vulnerabilities against the correct comparator. We label gene classes explicitly throughout because the enzyme set used for activity inference includes non-kinase signaling genes.

Materials and methods

Data sources (all open, no controlled access)

HCC proteome and phosphoproteome (159 tumors; 6,478 proteins, 26,418 phosphosites) came from the Gao 2019 supplementary tables (raw at NODE OEP000321) [1]. iCCA came from Dong 2022 (raw at NODE OEP001105) [2]; the cohort comprises 262 patients, of which 214 have proteome and phosphoproteome and 255 have mRNA, and we used the 214 with proteome and phosphoproteome for kinase and proteome analyses. TCGA-LIHC (371 primary tumors) and TCGA-CHOL (35 primary tumors) STAR log₂(TPM+1) and clinical data came from the UCSC Xena GDC hub [5]; we treat TCGA-CHOL as iCCA, noting it may include a minority of non-intrahepatic biliary cases. The cHCC-CCA cohort came from GEO GSE241466, a mixed series of 7 non-tumor, 5 HCC, 5 iCCA, and 8 intermediate-cell carcinoma samples; we used only the 8 i-CHC samples, identified from the series-matrix sample labels. For an independent cHCC-CCA replication we used GEO GSE231854 (25 cHCC-CCA and 56 HCC, RNA-seq), with labels from its series matrix. For a cross-organ generality test we used two single-cohort kidney proteomes from different laboratories, clear-cell RCC (Qu 2022; iProX IPX0001962000) and papillary RCC (Al Ahmad 2019; PRIDE PXD013523), with TCGA-KIRC and TCGA-KIRP RNA as the batch-free anchor. DepMap Public 24Q4 CRISPR (Chronos) and model metadata came from figshare [6,7]. Kinase-substrate relations came from OmniPath enzyme-substrate data, phosphorylation only (40,109 relations) [8,9]. Data sources are summarized in Table 1. We note that this enzyme universe includes non-kinase signaling genes such as adaptors, transporters, regulatory subunits, and pseudokinases, and we label gene classes explicitly rather than calling the whole set kinases.

Lineage axis: cross-validation, permutation, and non-marker control

TCGA Ensembl IDs were mapped to symbols (GENCODE v36) and collapsed by maximum; primary tumors were selected by barcode. On the 2,000 most variable genes (z-scored), a linear discriminant separated HCC from iCCA. Because an in-sample discriminant in a design with far more features than samples and a 371-versus-35 split separates trivially, we report a stratified 5-fold cross-validated AUC with a 200-fold label-permutation null. The cross-validation is fully nested: the 2,000-most-variable-gene selection (which is label-free) and the standardization are re-fit inside each training fold within a single pipeline, so the held-out fold informs neither the feature filter nor the scaler. We repeat the analysis after holding out the canonical hepatocyte and cholangiocyte marker panels to show the axis is not mere marker readout. For cross-dataset positioning we used a platform-robust single-sample score: within each sample, genes were percentile-ranked, and the score was the mean percentile of cholangiocyte markers (KRT19, KRT7, SOX9, EPCAM, CD24, SPP1, MUC1, CFTR, HNF1B, ANXA4, TACSTD2, CLDN4; we note SPP1 and CD24 are progenitor/secretory rather than strictly ductal markers) minus that of hepatocyte markers (ALB, APOA1, APOB, APOC3, TF, TTR, CYP3A4, HNF4A, FGA, FGB, FGG, SERPINC1, CPS1, ARG1). For i-CHC (n=8) we report Cliff’s delta effect sizes, exact Mann-Whitney P, and a bootstrap confidence interval for the median, and we note the dependence of the ordered-trend P on the large pure-lineage arms.

Identifiability test of the proteome and phosphoproteome, with a positive control

We did not treat unsupervised cohort separation as biology. For each shared protein we computed the iCCA minus HCC direction under within-sample percentile normalization (which cancels the global reference offset) and correlated it (Spearman, 1,000-fold bootstrap CI) against the batch-free

TCGA RNA direction; sign agreement on strongly directional genes (absolute RNA log2FC above 1) used a binomial test. As a non-aliased positive control, we computed the same direction in DepMap cell-line expression, where HCC and iCCA lines share one dataset so tumor type is not aliased with cohort, and correlated it with the TCGA direction. Per-sample kinase activities were estimated by kinase-substrate enrichment (KSEA; per-sample z of at least 5 substrate sites) [10,11] and benchmarked against the RNA axis. To show the negative correlation is harmonization-dependent rather than a fixed biological anti-correlation, we recomputed the proteome direction under four normalizations (raw difference, within-sample percentile, per-cohort z -score, quantile-merge). To make the failure concrete we ran a naive analyst’s pipeline (quantile normalization of the merged proteome, per-protein iCCA versus HCC Mann-Whitney with BH-FDR, validation against the RNA direction) and, because a competent analyst would use a dedicated corrector, ComBat [13] in two modes (batch set to cohort with no protected covariate, and with tumor type protected), reporting the discriminant separability, the fraction significant, and the anchor concordance for each. We then calibrated the standard batch-free anchor check with a ground-truth titration on TCGA-LIHC: a known 200-gene signature was spiked into a synthetic contrast aliased with a synthetic batch across 11 levels from 0 to 1 (20 repeats each), and at each level we measured recall, the false-discovery proportion, and the concordance of the recovered effect with a separate noisy orthogonal anchor; we report the pooled rank correlation with a level-block bootstrap (resampling the 11 levels) and the mean within-level association, under additive and multiplicative batch models. Finally, to test cross-organ generality, we ran the same naive harmonization and anchor validation on the kidney ccRCC versus pRCC pair.

Cell-line dependency with the correct comparator

In DepMap 24Q4, lines were classified by OncotreeCode into HCC ($n=24$), iCCA/IHCH ($n=31$), combined HCC and CC (KMCH1, $n=1$), and other ($n=1,122$); the later cell-line expression comparison (Figure 5B) used the 23 HCC and 30 iCCA lines that also have expression profiles. DepMap sample sizes therefore vary by data availability: the CRISPR dependency contrast uses 24 HCC and 31 iCCA lines, the line-expression comparisons use the 23 HCC and 30 iCCA lines with expression profiles, and program-level Chronos summaries use only lines with non-missing dependency values across the gene set. For each gene we computed selectivity versus all other lineages and the direct iCCA versus HCC contrast (Mann-Whitney with Benjamini-Hochberg FDR), and we annotated pan-lineage-essential genes (mean Chronos in other lineages below -0.5) and absolute essentiality (Chronos in iCCA lines below -0.5). TCGA differential expression used Mann-Whitney with BH-FDR. For the surviving signal we tested two gene programs, curated a priori from canonical pathway membership (iron handling; amino-acid transport) rather than derived from the contrast, at the program level by (i) a competitive permutation null (5,000 size-matched random gene sets) on the per-gene iCCA minus HCC dependency difference, (ii) a line-level Mann-Whitney of the per-line mean Chronos over the set (iCCA versus HCC lines), (iii) the same program’s coordinate expression in batch-free TCGA (CHOL versus LIHC), and (iv) the cross-line Spearman co-dependency structure. No multiplicity correction was applied across the small test family (two programs, four evidence legs each), so this analysis is exploratory. To quantify circularity, since two LAT1 genes are both in the program and among the selected genes, we repeated the dependency and expression tests after removing the discovery genes (SLC3A2, SLC7A5). Analyses used Python 3.13 (numpy, pandas, scipy, scikit-learn, statsmodels) and decoupleR with OmniPath [12]; no GPU was used. All scripts and intermediate tables are released at GitHub and Zenodo (DOI: 10.5281/zenodo.20888892).

Results

The HCC and iCCA lineage axis is robust and is not a marker artifact, and i-CHC is consistent with an intermediate phenotype

An in-sample discriminant separated HCC and iCCA almost perfectly (AUC 0.99985), but this is expected in a design with far more features than samples and is not used as evidence. Out of sample the lineage axis remained strong: the nested 5-fold cross-validated AUC was 0.966 (permutation null 95th percentile 0.629; permutation P below 0.005), and 0.966 when the canonical marker panels were held out, so the separation reflects broad transcriptomic lineage structure rather than a definitional readout of a few markers, and it is not an artifact of selecting features before cross-validation (Figure 1).

On the platform-robust single-sample score, i-CHC (n=8) fell between HCC and iCCA (medians -0.165, -0.082, +0.126; bootstrap CI for the i-CHC median -0.21 to 0.00). It was strongly separated from iCCA (Cliff's delta = -0.86, $P = 2.3\text{e-}5$) and modestly from HCC (delta = 0.41, $P = 0.045$). The limits matter. The ordered-trend statistic ($P = 4\text{e-}23$) is dominated by the large, perfectly separated pure-lineage arms and does not by itself establish i-CHC intermediacy. The i-CHC versus HCC contrast rests on 8 samples and a marginal P. An earlier analysis that mistakenly included the cohort's non-i-CHC samples flattened the result toward HCC, and correcting the cohort label changed the conclusion, which shows the fragility of small-sample cross-dataset positioning. We therefore frame i-CHC as consistent with an intermediate transcriptomic phenotype. An independent cohort supports the direction of this result: in GSE231854 (25 cHCC-CCA and 56 HCC, a separate study), cHCC-CCA was significantly shifted toward the cholangiocyte pole relative to HCC (Cliff's delta = 0.58, one-sided $P = 1.7\text{e-}5$; Figure 5A). GSE231854 contains no pure-iCCA arm, so it replicates only the cholangiocyte-ward shift relative to HCC, not the full intermediacy. The intermediacy between both poles therefore rests on the primary 8-sample i-CHC cohort, which alone spans HCC and iCCA, while GSE231854 confirms the HCC versus cHCC-CCA direction in a larger, independent sample.

Single-cohort proteome and phosphoproteome do not recover lineage biology, an identifiability result

The two open proteogenomic cohorts share 6,067 proteins and 14,246 phosphosites, which superficially licenses a manifold. They do not support one. Per-cohort z-scoring removed the lineage-marker contrast (marker-score discriminant AUC 0.50), the arithmetic consequence of normalizing away a mean shift that is wholly between-cohort; a fully multivariate classifier still separates the cohorts after the same z-scoring (cross-validated AUC 0.975), but that residual separability is the batch artifact we decline to read as biology rather than recovered lineage signal. Within-sample percentile normalization, which preserves internal ordering, yielded a proteome iCCA minus HCC direction that was significantly discordant with the batch-free RNA direction (Spearman rho = -0.15, 95% CI -0.18 to -0.13; sign agreement 42%, binomial $P = 2\text{e-}12$; Figure 2). The phosphoproteome was uninformative: per-sample KSEA scored 237 kinases in both cohorts and flagged 134 with nominally significant cross-tumor activity differences, but these carried no lineage information (rho = -0.045; sign agreement 45%, binomial $P = 0.17$).

A non-aliased positive control calibrates how much concordance is even achievable: the iCCA minus HCC direction in DepMap cell-line expression, where HCC and iCCA lines share one dataset so tumor type is not aliased with cohort, correlated with the TCGA direction only negligibly (rho = 0.015, $P = 0.041$, n = 19,193). The achievable ceiling for per-gene cross-dataset concordance

therefore sits far below any useful threshold. The observed proteome correlation is not near zero but mildly negative ($\rho = -0.15$); rather than name a mechanism, we show it is harmonization-dependent, ranging across four normalizations from $\rho = -0.21$ (raw) through -0.15 (within-sample percentile) and -0.13 (quantile-merge) to -0.004 (per-cohort z-score). The sign moves with the normalization choice and, given the near-zero ceiling, is within ordinary cross-modality discordance and uninterpretable as biology. These data cannot support a trustworthy cross-tumor axis, because the achievable signal sits below the noise floor of aliasing.

To make this concrete, we ran the pipeline a naive analyst would use. Quantile harmonization of the merged Gao-HCC and Dong-iCCA proteome, followed by a per-protein iCCA versus HCC test, flagged 3,796 of 6,067 proteins (63%) as significant at FDR below 0.05, among them 407 named signaling enzymes. Yet these hits did not validate against the batch-free RNA axis (Spearman $\rho = -0.15$, $P = 2e-19$; sign agreement 40%): a standard harmonization does not merely fail to find signal, it manufactures thousands of plausible false positives that a batch-free check exposes as artifacts. A dedicated corrector does not escape the trap, as the identifiability argument predicts. ComBat with batch set to cohort and no protected covariate erased the contrast (discriminant AUC 0.50, no proteins significant, anchor $\rho = -0.13$), exactly like per-cohort z-scoring, because the only between-cohort signal it can remove is the tumor-type contrast itself. Protecting tumor type instead is ill-posed under perfect aliasing: the design pairing cohort batch with the tumor-type covariate is rank-deficient (rank 2 of 3 columns), the configuration the confounded-design guards in standard implementations (sva and pyComBat) reject. Forced through the pseudo-inverse it does not recover the axis (the cohorts no longer separate, AUC 0.51), and its direction shows a spuriously positive anchor concordance ($\rho = +0.18$, above any honest normalization and the $+0.015$ ceiling): not recovered biology but the protected covariate being re-imprinted, the aliased artifact itself. The tool’s own guard thus operationalizes the non-identifiability, and is the reason we route every cross-tumor direction through a batch-controlled layer.

Calibrating a standard anchor check for the single-cohort setting

Validating a cross-cohort contrast against a batch-free reference and distrusting low concordance is established practice [21,22,23,24]; our contribution is to calibrate how that check degrades with aliasing where it is usually skipped. In a ground-truth titration on TCGA-LIHC expression, a known 200-gene signature was spiked into a synthetic contrast whose aliasing with a synthetic batch was swept from 0 to 1 (20 repeats per level), and concordance was read against a separate noisy orthogonal anchor (concordance to the truth about 0.20; Figure S1). As aliasing rose, true-signal recall stayed high (above 0.82) but the false-discovery proportion climbed from 0.06 to 0.93, and the noisy anchor tracked this across regimes (pooled Spearman -0.68 , level-block bootstrap 95% CI -0.72 to -0.43 ; -0.68 under a multiplicative model). Two bounds apply: the per-level shift in anchor concordance is small (0.105 to 0.042), and within a fixed aliasing level the anchor-versus-false-discovery association is essentially nil (mean within-level Spearman -0.02), because the anchor shares the injected signal with the recovered effect by construction. The titration therefore calibrates sensitivity to aliasing regimes, not an independent per-replicate score. The failure recurs in kidney: naively harmonizing two single-cohort, different-lab kidney proteomes (clear-cell versus papillary RCC) flagged 6,642 of 8,787 proteins (76%) as different, none correlating with the batch-free TCGA-KIRC versus KIRP anchor ($\rho = 0.004$).

Apparent iCCA-selective dependencies are largely a comparator artifact, and only a small amino-acid-transport signal survives a fair test

A naive selectivity definition, lower Chronos in iCCA lines than in all other lineages, at first glance nominated 11 iCCA-selective kinase dependencies, of which 9 appeared iCCA-up in batch-free RNA, an apparently coherent receptor-tyrosine-kinase, MAPK, and amino-acid-transport program (Figure 3). This artifactual positive, a version of which was the headline of our own first draft, does not survive scrutiny. The all-other-lineages comparator is overwhelmingly non-liver, so it measures liver versus non-liver essentiality, not HCC versus iCCA specificity. Under the correct direct iCCA versus HCC contrast with FDR control, only 2 of the 11 survived (SLC3A2 and TRPM7); 7 were pan-lineage-essential (for example CDK1, with Chronos near -2.5 in both liver lineages), and 3 (GRB2, CCND1, CRKL) were flagged selective for both liver lineages, that is liver-essential rather than iCCA-specific. EGFR, the most clinically suggestive hit, was neither RNA-concordant by the prespecified threshold ($\log_2\text{FC}$ 0.38, below 0.5) nor differentially essential between the liver lineages (raw Mann-Whitney $P = 0.15$, BH-FDR 0.63), and its absolute dependency was marginal (Chronos -0.56). Several genes in the apparent set are not kinases at all: SLC7A5 and SLC3A2 are the LAT1 transporter, GRB2 and CRKL are adaptors, CCND1 is a cyclin, PPP1R12A is a phosphatase subunit, and ILK is a pseudokinase, so the kinase framing is itself inaccurate.

The direct contrast (FDR below 0.1) independently flags four genes, TFRC (transferrin receptor), SLC3A2 (LAT1 amino-acid-transport subunit), TRPM7 (an ion-channel kinase), and NHEJ1 (Table S1), comprising the two retained from the all-other-lineage set (SLC3A2, TRPM7) plus TFRC and NHEJ1, which were not in it. Three of the four (TFRC, SLC3A2, and the channel-kinase TRPM7) point to nutrient and ion transport rather than the textbook RTK axis; NHEJ1, a DNA-repair factor, has no transport rationale and is most likely an incidental survivor, which itself argues for caution about reading a four-gene set as a program. We do not present these as validated targets ($n = 31$ versus 24; FDR 0.04 to 0.08; TFRC and TRPM7 are preferential but pan-essential). Separately, FGFR1 illustrates that expression is an unreliable dependency proxy: its mRNA is markedly iCCA-higher ($\log_2\text{FC} +3.16$) yet its dependency is higher in HCC lines. FGFR1 is not the actionable iCCA target; the clinically validated program is FGFR2 fusions and IDH1 mutations [4,15], neither captured by bulk mRNA or unselected DepMap lines.

A corroborated positive, amino-acid transport (LAT1), recovered by the comparator-correct pipeline

The one program-level signal that is corroborated, rather than dissolved, under further testing is amino-acid transport, though its legs differ in strength and in their independence from the gene selection, which we are careful to separate. We tested two pathway programs, curated a priori from canonical membership (iron handling; amino-acid transport) rather than derived from the contrast, at the program level (Figure 4); no formal multiplicity correction was applied across the programs and axes, so this analysis is exploratory hypothesis-generation. The iron-handling program did not generalize: it was not preferentially essential in iCCA lines (competitive permutation $P = 0.14$) and not coordinately expressed (batch-free TCGA $P = 0.55$), so the single-gene TFRC hit does not reflect a broader iron-addiction phenotype.

For the amino-acid-transport program, the three legs are not equally independent of the gene selection, and we weight them accordingly. The load-bearing, selection-independent evidence is expression: the program is strongly and coordinately over-expressed in iCCA tumors (TCGA-CHOL versus LIHC: median $\log_2\text{FC} +0.79$; 86% of genes iCCA-up; set-score $P = 1.2\text{e-}10$), and

this survives removing the two discovery genes, because dropping SLC3A2 and SLC7A5 leaves $P = 1.6\text{e-}10$ with 83% of remaining genes iCCA-up, so it is not an artifact of the genes that defined the signal. The CRISPR-dependency leg, in contrast, is partly circular: SLC3A2 and SLC7A5 (the LAT1 subunits) were themselves among the selected genes, and the program’s apparent iCCA-preferential dependency (competitive permutation $P = 0.0024$) is carried by them, since removing both collapses it to $P = 0.20$ (set-mean delta Chronos -0.059 to -0.011); both the full-set and leave-out values now come from a single seeded run over a common background. The selection-independent dependency evidence is therefore only the marginal line-level test (Mann-Whitney $P = 0.045$ for the amino-acid-transport set; iCCA mean Chronos -0.23 versus HCC -0.17, both above the -0.5 essentiality floor), and the co-dependency evidence (TFRC and SLC3A2 are co-dependent, $\rho = 0.56$, though the broader set is not, mean ρ 0.11; Figure 4C) likewise shares genes with the selection. We thus rest the program-level claim on the selection-independent expression axis plus the clinical anchor, and present the lineage-preferential dependency as the study’s own, more modest observation rather than a strong addiction. The over-expression is a property of iCCA tumors rather than of iCCA cell lines: in DepMap line expression the amino-acid-transport program was not elevated in iCCA versus HCC lines (one-sided $P = 0.35$, on the 23 HCC and 30 iCCA lines with expression data; the effect was near zero and if anything slightly lower in iCCA; Figure 5B). We read this as tumor-level or differentiation-dependent biology that passaged lines lose, which is consistent with the clinical evidence that LAT1 marks patient tumors and outcomes rather than cell lines, and it explains why the dependency signal in lines is weak.

The clinical anchor is specific to the transporter our analysis flags. The LAT1 (SLC7A5/SLC3A2) inhibitor nanvuranlat (JPH203) improved progression-free survival in a randomized, placebo-controlled phase 2 trial in advanced biliary tract cancer (hazard ratio 0.56, 95% CI 0.34 to 0.90, $P = 0.02$) [16], and high LAT1 expression is an established poor-prognosis marker in biliary tract cancer [17,18]. We report LAT1 and amino-acid transport not as a new discovery, but as the lineage-preferential dependency that survives the comparator-correct pipeline while the EGFR artifact falls away. The limits remain: the dependency effect is modest and partly circular, the nanvuranlat result is a phase 2 PFS signal (not overall survival or phase 3) in biliary tract cancer broadly rather than iCCA alone, and the over-expression is measured in unadjusted bulk tumors, so part of the differential could reflect grade or proliferation rather than lineage alone (see Limitations).

Discussion

We tested whether open liver cancer proteogenomic data support a cross-tumor kinase-activity manifold, and what they can support instead. Two contributions are robust. First, we provide a cross-validated, marker-independent HCC and iCCA lineage axis on which rare intermediate-cell cHCC-CCA is consistent with an intermediate phenotype, with the caveat that the primary cHCC-CCA evidence rests on eight RNA-only samples. Second, we provide a cautionary benchmark: when tumor type is perfectly aliased with cohort, the proteome and phosphoproteome cannot be combined into a trustworthy cross-tumor axis. This is a design-level non-identifiability problem, not a normalization problem.

The functional-genomics result makes the same informatics point in a different layer: selectivity is only as meaningful as its comparator. Defining iCCA-selective against all other lineages produced a coherent-looking EGFR/GRB2/CRKL program that mostly evaporated under a direct iCCA versus HCC test, a common way to mistake pan-essential or lineage-general genes for tumor-type-specific dependencies. The comparator-correct procedure also recovers a constructive result:

amino-acid transport (LAT1: SLC7A5/SLC3A2, with glutaminolysis and TFRC) is supported by selection-independent tumor expression and by the phase 2 nanvuranlat PFS signal [16]. Its iCCA-preferential CRISPR dependency remains modest and partly circular. The iron-handling program, despite a strong TFRC single-gene hit, did not survive as a program, reinforcing the need to test pathways rather than narrate them from single genes.

For practitioners the guidance is concrete: anchor every cross-tumor direction to a batch-free layer; treat single-cohort-per-type proteogenomic merges as non-identifiable for the between-type contrast; cross-validate supervised separations when features greatly outnumber samples; and define dependency selectivity against the biologically relevant comparator with multiple-testing control. The same trap appears in kidney (ccRCC versus pRCC), and the ground-truth calibration shows the anchor check degrades with aliasing across regimes. The anchor check itself is established practice [21,22,23,24]; the contribution here is the calibration, the non-identifiability framing for this single-cohort-per-type case, and a transparent benchmark in a powered liver case with a corroborating, unpowered kidney case. We frame this as a prospective caution: we know of no published study that has merged these specific cohorts across tumor type, and the benchmark is meant to forestall a tempting analysis rather than rebut a committed one.

Limitations

1. Aliasing of tumor type with cohort (and, for HCC, HBV etiology) is intrinsic to the open data and is the dominant limitation; a definitive proteogenomic axis would need a second independent cohort per tumor type, which is not openly available.
2. Comparison groups are small: TCGA-CHOL (n=35), i-CHC (n=8, intermediacy preliminary though its direction replicated in GSE231854), and DepMap iCCA (n=31) versus HCC (n=24); IHCH annotation may be impure.
3. cHCC-CCA is RNA-only in open data, so kinase activity cannot be measured there, and the intermediate-cell i-CHC subtype is uncommon and may not represent the broader cHCC-CCA spectrum (the GSE231854 replication is unsubtyped).
4. The exploratory iCCA-preferential set (TFRC, SLC3A2, TRPM7, NHEJ1) rests on four genes at FDR 0.04 to 0.08 and is a hypothesis, not a result; NHEJ1 (delta Chronos -0.16, non-essential in both arms) should not be read at the status of TFRC or SLC3A2.
5. KSEA and the OmniPath network are biased toward well-studied genes, and cross-cohort KSEA suffers differential phosphosite coverage.
6. The lineage score and LAT1 expression test use unadjusted bulk tumors (TCGA-LIHC is etiologically mixed; grade and proliferation were not adjusted), and the two-program transport analysis applies no cross-program multiplicity correction, so it is exploratory.

Conclusion

Open, batch-controlled transcriptomic data define a robust HCC and iCCA lineage axis and position intermediate-cell cHCC-CCA as consistent with an intermediate phenotype, while the full position between both poles still rests on eight primary RNA samples. Single-cohort proteome and phosphoproteome data cannot support a cross-tumor kinase-activity manifold because the contrast is non-identifiable under perfect aliasing. Apparent lineage-selective dependencies likewise require the correct comparator. Tested fairly, the LAT1 amino-acid-transport program remains a modest but clinically anchored positive result, whereas the larger kinase program dissolves. We release the pipeline as a reusable cautionary benchmark for liver-cancer multi-omics integration.

Data and code availability

All data are public (NODE OEP000321; NODE OEP001105; UCSC Xena GDC hub for TCGA-LIHC and TCGA-CHOL; GEO GSE241466; GEO GSE231854; iProX IPX0001962000; PRIDE PXD013523; TCGA-KIRC and TCGA-KIRP; DepMap 24Q4; OmniPath). The complete pipeline, result tables, figures, manuscript build scripts, and parsed intermediate matrices are released at GitHub (<https://github.com/Goden26/single-cohort-proteogenomic-nonidentifiability>) and archived at Zenodo (DOI: 10.5281/zenodo.20888892). Raw public input files are not redistributed in the code repository; `data_manifest.md` and the download scripts list the public sources and expected filenames. The environment is Python 3.13 with numpy 2.4, pandas 2.3, scipy 1.18, scikit-learn 1.9, statsmodels 0.14, decoupleR 2.1.6, and omnipath 1.0.12; the OmniPath kinase-substrate snapshot used here returned 40,109 phosphorylation relations and 1,657 enzymes (retrieved 2026-06). A top-level script regenerates the stat files, tables, and figures.

Declarations

Ethics approval. Not applicable. The study used only public, previously published, de-identified or aggregate datasets; no new human or animal data were collected.

Competing interests. The author declares no competing interests.

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CRedit author contributions. Anh Tuan Quan: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review and editing. The author is the sole contributor and holds all applicable roles.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used an AI-assisted coding and writing tool (Anthropic Claude, accessed through Claude Code) to help locate and download public datasets, write and debug analysis code, draft and revise text, and run an internal simulated peer review. After using this tool, the author reviewed and edited the content as needed, specified every analysis, verified that all reported statistics were produced by the released scripts, independently checked that all cited references are real and correctly attributed, and takes full responsibility for the content and conclusions.

Figure legends

Figure 1. A cross-validated, marker-independent HCC and iCCA lineage axis with cHCC-CCA positioned intermediate. Left: PCA of TCGA-LIHC (HCC, n=371) and TCGA-CHOL (iCCA, n=35) primary tumors (identical GDC pipeline); nested cross-validated AUC was 0.966 (0.966 with markers held out). Right: cholangiocyte minus hepatocyte single-sample score across HCC, intermediate-cell i-CHC (GSE241466, n=8) and iCCA; i-CHC is strongly separated from iCCA (delta = -0.86, P = 2.3e-5) and modestly from HCC (delta = 0.41, P = 0.045, with the bootstrap median CI touching zero), so the HCC-side separation is suggestive at n=8 rather than definitive.

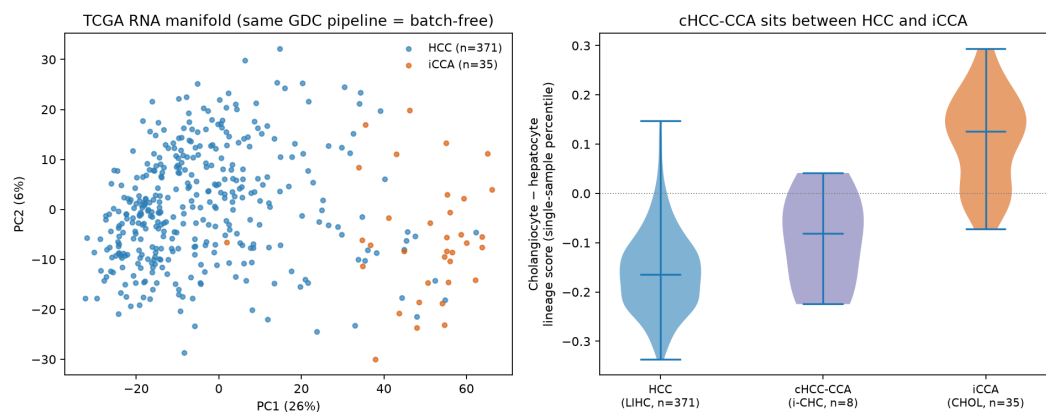


Figure 2. Single-cohort proteome data fail to recover the batch-free lineage axis. Left: PCA of within-sample-normalized Gao-HCC and Dong-iCCA proteomes; samples separate by cohort, which is perfectly aliased with tumor type. Right: per-protein iCCA minus HCC direction (proteome) versus batch-free TCGA RNA direction, which is significantly discordant (Spearman $\rho = -0.15$, 95% CI -0.18 to -0.13).

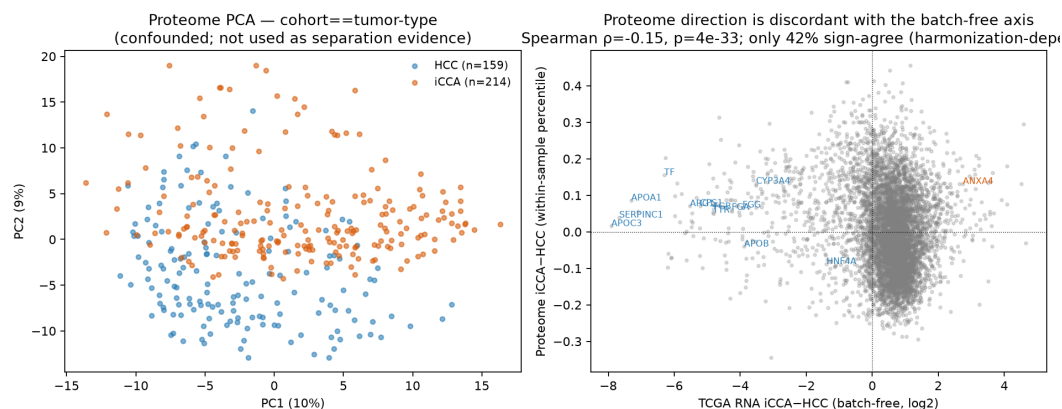


Figure 3. Apparent iCCA-selective dependencies and what survives a fair test. DepMap iCCA-selectivity versus all-other-lineages (left) nominates 11 hits, but the direct iCCA versus HCC contrast with FDR (right) leaves only TFRC, SLC3A2, TRPM7, NHEJ1; pan-essential and selective-for-both genes are flagged.

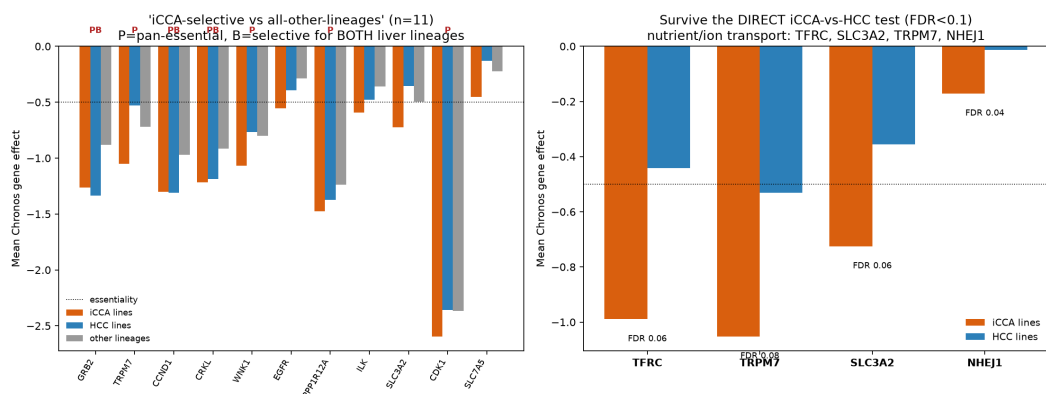


Figure 4. The amino-acid-transport (LAT1) program is preferentially essential and over-expressed in iCCA. Left: per-gene iCCA minus HCC dependency difference for the iron and amino-acid-transport genes (orange = more essential in iCCA); TFRC, SLC3A2, SLC7A5, GLS lead. Middle: program-level Chronos over the combined iron and amino-acid-transport set, iCCA versus HCC lines (combined set: line-level Mann-Whitney $P = 0.096$, competitive permutation $P = 0.0020$; amino-acid-transport set alone: Mann-Whitney $P = 0.045$, permutation $P = 0.0024$). The dependency permutation signal does not survive removing the LAT1 discovery genes SLC3A2 and SLC7A5 ($P = 0.20$), so it is partly circular; the batch-free expression corroboration does survive it ($P = 1.6e-10$, 83% of remaining genes iCCA-up). Right: cross-line co-dependency (Spearman) among iron and amino-acid-transport genes; TFRC and SLC3A2 are co-dependent ($\rho = 0.56$), although the broader set is not coherently co-dependent (mean ρ 0.11). The iron-handling program alone did not generalize (permutation $P = 0.14$; expression $P = 0.55$).

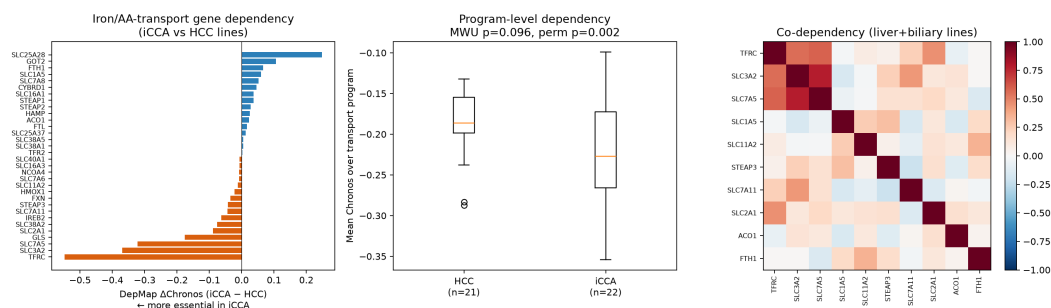
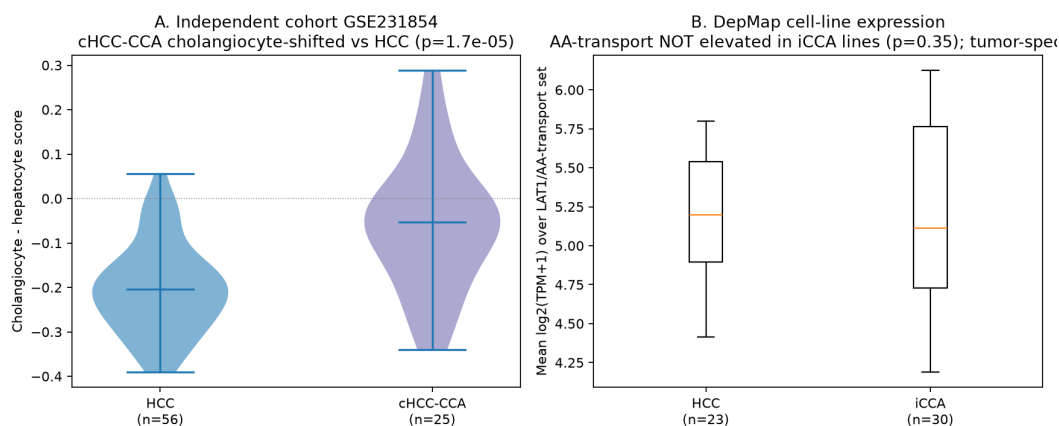


Figure 5. Independent replications. (A) In an independent cohort (GSE231854; 25 cHCC-CCA and 56 HCC), cHCC-CCA was significantly shifted toward the cholangiocyte pole relative to HCC on the single-sample lineage score (Cliff's delta = 0.58, one-sided $P = 1.7e-5$), which replicates the direction of Figure 1. (B) In DepMap cell-line expression, the LAT1 and amino-acid-transport program was not elevated in iCCA versus HCC lines (one-sided $P = 0.35$), which indicates that the over-expression seen in tumors (Figure 4) is tumor-specific.



Tables

Table 1. Datasets used. All are public and openly downloadable.

Cohort	Source or accession	N used	Data type	Role
Gao 2019 HCC	NODE OEP000321	159 tumors	proteome, phosphosite	proteome and phospho
Dong 2022 iCCA	NODE OEP001105	214 proteome and phospho; 255 mRNA	proteome, phosphosite, mRNA	proteome and phospho
TCGA-LIHC	UCSC Xena GDC hub	371 tumors	RNA-seq	batch-free lineage axis
TCGA-CHOL	UCSC Xena GDC hub	35 tumors	RNA-seq	batch-free lineage axis
GSE241466 (i-CHC)	GEO	8 i-CHC	RNA-seq	cHCC-CCA positioning
GSE231854	GEO	25 cHCC-CCA, 56 HCC	RNA-seq	independent cHCC-CCA replication
DepMap 24Q4	figshare	24 HCC, 31 iCCA lines	CRISPR Chronos and expression	dependency and line expression
ccRCC proteome	iProX IPX0001962000 (Qu 2022)	232 tumors	proteome	kidney generality
pRCC proteome	PRIDE PXD013523 (Al Ahmad 2019)	16 tumors	proteome	kidney generality
TCGA-KIRC, TCGA-KIRP	UCSC Xena GDC hub	537, 290 tumors	RNA-seq	kidney batch-free anchor
OmniPath	Enzsub	1,657 enzymes	kinase-substrate network	KSEA reference

Supplementary material

Figure S1. Ground-truth calibration of the anchor check. Using real TCGA-LIHC expression as substrate, a known 200-gene signature was spiked into a synthetic biological contrast, a synthetic batch effect was added to a synthetic cohort, and the aliasing of the contrast with the cohort was swept across 11 levels from 0 to 1 (20 repeats per level). Concordance is measured against a separate noisy orthogonal anchor (concordance to the truth only about 0.20), not the ground truth. Left: as aliasing increases, true-signal recall stays high (blue) while the false-discovery proportion rises to 0.93 (orange) and the noisy-anchor concordance shifts only modestly (dashed, 0.105 to 0.042). Right: pooling the 220 runs, the noisy-anchor concordance predicts the false-discovery proportion (Spearman -0.68, level-block bootstrap 95% CI -0.72 to -0.43; -0.68 under a multiplicative batch model). The within-level (alpha-conditioned) association is near zero (mean Spearman -0.02), and the anchor shares the injected signal with the recovered effect by construction, so the calibration reflects sensitivity to the aliasing regime, not an independent per-replicate association.

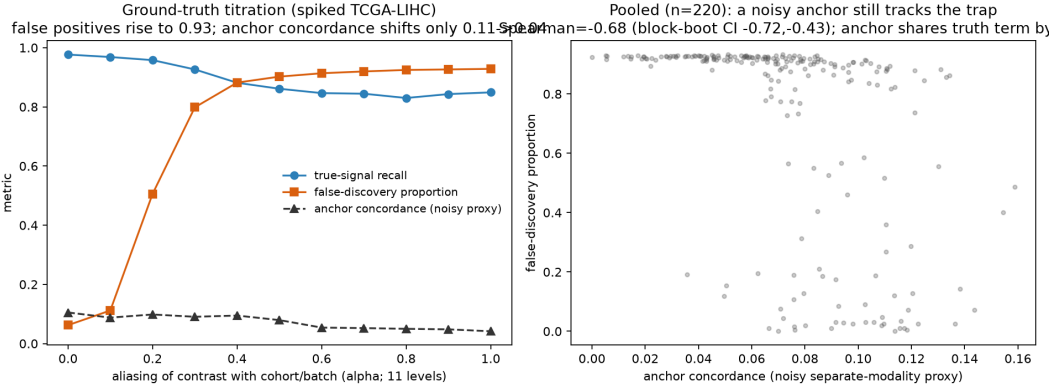


Table S1. Genes passing the direct iCCA versus HCC dependency contrast (FDR below 0.1). Negative dChronos means more essential in iCCA lines. TFRC and TRPM7 remain pan-essential, so they are preferential rather than iCCA-exclusive. FDR reflects rank separation rather than effect magnitude, so NHEJ1 can show a low FDR despite a small dChronos; NHEJ1 has no transport rationale and is likely an incidental survivor.

Gene	dChronos (iCCA minus HCC)	FDR	Chronos iCCA	Chronos HCC	RNA log2FC (iCCA minus HCC)
TFRC	-0.548	0.063	-0.989	-0.441	0.642
TRPM7	-0.521	0.075	-1.052	-0.531	0.633
SLC3A2	-0.369	0.063	-0.725	-0.355	0.759
NHEJ1	-0.159	0.045	-0.171	-0.012	0.460

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Declarations

Ethics approval and consent to participate

Not applicable. The study used only publicly available, previously published, de-identified or aggregate datasets (TCGA, DepMap, GEO, NODE-deposited proteogenomic cohorts, and OmniPath). No new human or animal data were collected, and no identifiable participant data were used.

Consent for publication

Not applicable.

Declaration of competing interest

The author declares no competing interests.

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CRedit author contributions statement

Anh Tuan Quan: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review and editing. The author is the sole contributor and holds all applicable roles.

Availability of data and materials

All datasets analyzed are public and openly accessible: NODE OEP000321 (Gao 2019 HCC proteogenomics) and NODE OEP001105 (Dong 2022 iCCA proteogenomics), with processed matrices in the respective journal supplementary tables; TCGA-LIHC, TCGA-CHOL, TCGA-KIRC, and TCGA-KIRP via the UCSC Xena GDC hub; GEO GSE241466 and GSE231854; the clear-cell renal cell carcinoma proteome (iProX IPX0001962000) and the papillary renal cell carcinoma proteome (PRIDE PXD013523); DepMap Public 24Q4 via figshare; and the OmniPath kinase-substrate network. The complete analysis code (Python 3.13), parsed intermediate matrices, per-gene result tables, figure-generation scripts, and manuscript build scripts are released at <https://github.com/Goden26/single-cohort-proteogenomic-nonidentifiability> and archived at Zenodo (DOI: 10.5281/zenodo.20888892); a single top-level script regenerates the stat files, tables, and figures.

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None.

Use of generative AI and AI-assisted technologies (disclosure)

During the preparation of this work the author used an AI-assisted coding and writing tool (Anthropic Claude, accessed through the Claude Code interface) to help locate and download public datasets, to write and debug the Python analysis code, to draft and revise the manuscript text, and to run an internal simulated peer review that guided the analysis and the writing. After using

this tool the author reviewed and edited the content as needed, directed every analysis, verified that all reported statistics were produced by the released code, independently confirmed that all 24 cited references are real and correctly attributed (PubMed identifiers are listed), and takes full responsibility for the content, the interpretation, and the conclusions. No data, results, or citations were generated without verification against the primary source or the released pipeline.